

Dry-stone walls

Spatial scale: vineyard / landscape

About this sustainable management strategy

Dry-stone walls are traditional masonry structures built without binders such as mortar or cement, using locally sourced stones often collected from cultivated land or nearby quarries. Their construction relies on a foundation of large blocks, successive layers of coarser material, and smaller stones filling the gaps to ensure stability, with wall dimensions adapted to slope conditions and typically tapering towards the top. The dry laying allows rainwater drainage and is often combined with a crushed stone backing to reduce hydrostatic pressure.

Dry-stone walls are integral components of terraced and agricultural landscapes, shaped over centuries through the interaction between human management and natural processes. They are considered "living" structures due to their ecological functions and interactions within the environment. Along with other structural elements such as hedgerows and stone heaps, dry-stone walls, create valuable ecological niches in agricultural landscapes, particularly in viticultural areas.

Key ecosystem services

- Erosion control (*slope stabilisation*)
- Water retention (*reduced surface runoff*)
- Microclimate regulation (*reduced temperature extremes*)
- Water regulation (*improved infiltration, retention, and reduced nutrient leaching*)
- Habitat provision and biodiversity conservation (*microhabitats for plants, invertebrates, reptiles, and small mammals*)
- Cultural heritage (*preservation of traditional knowledge and historical land-use practices*)
- Landscape aesthetics (*distinct landscape character and enhanced recreational value*)

Considerations for successful implementation

Construction and maintenance expertise

The long-term stability and functionality of dry-stone walls depend on appropriate construction techniques and periodic maintenance. Preserving and transferring traditional craftsmanship plays an important role in ensuring the durability and ecological value of these structures.

Availability of skills and materials

The restoration and maintenance of dry-stone walls depend on access to appropriate stone materials and specialised construction knowledge. In some regions, the availability of skilled craftspeople and suitable materials may influence the feasibility and long-term maintenance of these structures.

Long-term conservation and landscape management

The ecological and cultural benefits of dry-stone walls are maintained through continued management and integration within broader landscape conservation efforts. Preventing abandonment and supporting the maintenance of traditional vineyard landscapes contribute to the preservation of biodiversity, cultural heritage, and ecosystem functions.